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Roll No.

THIRD SEMESTER--B.TECH- Course Work
END SEMESTER EXAMINATION, NOV-DEC, 2023

Course Code: ICMTTC04/EEMTC04

Course Title: Mathematics III

Time- 3:00 hours

Max. Marks- 50

Note:- Attempt all the five questions. Missing data/information (if any), may be suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
Q1.	Attempt any 2 parts of the following		
1a	Show that the polar form of Cauchy Riemann equations are (a) $\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}$, $\frac{\partial u}{\partial \theta} = -r \frac{\partial v}{\partial r}$ (b) Deduce that $\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0$	5	CO1
1b	Find the image and draw a rough sketch of the mapping of the region $1 \leq x \leq 2$ and $2 \leq y \leq 3$ under the mapping $w = e^z$	5	CO1
1c	Evaluate $\int_{-\infty}^{\infty} \frac{1}{x^4 + 1} dx$	5	CO1
Q2.	Attempt any 2 parts of the following		
2a	Find $\nabla \phi$ and $ \nabla \phi $ when $\phi = (x^2 + y^2 + z^2)e^{-(x^2 + y^2 + z^2)^{1/2}}$	5	CO2
2b	Find the circulation of $\vec{F}$ around the curve C, where $\vec{F} = (e^x \sin y)i + (e^x \cos y)j$ and C is the rectangle whose vertices are $(0,0)$; $(1,0)$; $(1,\pi/2)$; $(0,\pi/2)$	5	CO2
2c	If $\vec{F} = 2yi - zj + x^2k$ and S is the surface of the parabolic cylinder $y^2 = 8x$ in the first octant bounded by the planes $y = 4$ and $z = 6$ Evaluate $\iint_S \vec{F} \cdot \hat{n} ds$	5	CO2
Q3.	Attempt any 2 parts of the following		
3a	Using Laplace Transform evaluate $\int_0^{\infty} e^{-2t} \sin^3 t dt$	5	CO3
3b	State and prove Convolution Theorem	5	CO3

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3c	Find the extremals of the functional and extremum value of the following $I[y(x)] = \int_{x_0}^{x_1} \frac{1+y^2}{y'^2} dx$	5	CO3
Q.4	Attempt any 2 parts of the following		
4a	Find the half range sine series for the function $f(x) = e^x$, for $0 < x < 1$	5	CO4
4b	Find the Fourier series for the function $f(x) = x + x^2$ in the interval $-\pi < x < \pi$	5	CO4
4c	A periodic function of period 4 is defined as $f(x) = x , -2 < x < 2.$ Find its Fourier series expansion.	5	CO4
Q5.	Attempt any 2 parts of the following		
5a	A bar with insulated sides is initially at temperature 0°C throughout. The end $x = 0$ is kept at 0°C for all time and the heat is suddenly applied so that $\frac{\partial u}{\partial x} = 10$ and $x = l$ for all time. Find the temperature function $u(x, t)$	5	CO5
5b	A string is stretched and fastened to two points l apart. Motion is started by displacing the string in the form $u = a \sin \frac{\pi x}{l}$ from which it is released at time $t=0$. Find the displacement $u(x, t)$ from one end	5	CO5
5c	Solve the equation $\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = 0$ with conditions (i) $V = 0$ when $y \rightarrow \infty$ for all x (ii) $V = 0$ when $x = 0$ for all values of y (iii) $V = 0$ when $x = 1$ for all values of y (iv) $V = x(1-x)$ when $y=0$ for $0 < x < 1$	5	CO5